Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

DevOps Foundation Training

TGS Ref No: TGS-2024049184

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 4.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	Legacy release	Legacy slide release retained as source evidence.	Dr. Alfred Ang
4.0	20 August 2026	Two-day 16-hour plan aligned to final v4.0 slide numbers, eight R labs, corrected K1–K5/A1–A6 mapping, and the approved 60-minute WA plus 60-minute Case Study.	Dr. Alfred Ang

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Course Information

Field	Details
Course	DevOps Foundation Training (TGS-2024049184)
TSC	Problem Management (ICT-OUS-4011-1.1)
Training provider	Tertiary Infotech Academy Pte Ltd · UEN: 201200696W
Duration	2 days · 16 hours total · 14 hours classroom facilitation + 2 hours assessment
Daily timing	09:00–18:00 with one-hour lunch excluded; short comfort breaks are integrated into facilitated time
Mode	Instructor-led classroom learning with peer sharing, group discussion, case-based analysis and self-contained R labs
Assessment	WA-SAQ, 60 minutes, open book + Case Study, 60 minutes, open book
Trainer	Dr. Alfred Ang

Learning Outcomes and TSC Mapping

LO	Learning outcome	Knowledge	Abilities
LO1	Manage the lifecycle of problems using DevOps principles and technologies.	K1, K2	A1, A2
LO2	Perform investigations and develop DevOps methods for problem analysis.	K3, K4	A3, A4
LO3	Recommend DevOps solutions and monitor documentation for problem solving.	K5	A5, A6

Knowledge and Ability Statements

Code	Statement
K1	Principles of problem management throughout its lifecycle
K2	Relevant tools, processes and technologies for problem identification, investigation, analysis and resolution
K3	Problem investigation and diagnosis techniques and methodologies
K4	Problem prioritisation and sizing techniques, methodologies and parameters
K5	Best practices and industry standards in problem-management documentation
A1	Manage the lifecycle of a wide range of problems
A2	Introduce technologies and processes for automated detection of incidents or problems
A3	Investigate and deeply analyse a problem to understand its root causes
A4	Develop guidelines for prioritising and categorising problems by severity, frequency and implications
A5	Recommend solutions that address root causes and minimise recurrence
A6	Monitor documentation and tracking of problems encountered and resolved

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Instructional and Assessment Strategy

Method	How it is used	Evidence
Concept-led facilitation	Editable diagrams, comparisons and native PowerPoint charts explain systems and trade-offs without click-by-click procedure.	Retrieval prompts, explanations and concept checks
Peer sharing	Learners compare workplace signals, constraints and cultural conditions in groups of up to five.	Evidence-based observation and rationale
Group discussion	Teams challenge hypotheses, scoring policies and recommendations against supplied evidence.	Documented trade-off and decision
R labs	Eight self-contained scripts turn synthetic incident, pipeline and metric data into auditable outputs.	CSV, PNG, Markdown and checklist evidence
WA-SAQ	Five open-ended items test K1–K5 from the taught topics.	Written responses; 60 minutes
Case Study	Three connected tasks test A1–A6 using the same problem-management storyline practised in the labs.	Analysis and recommendations; 60 minutes

Course Schedule

Day 1 — DevOps foundations, flow, feedback, practices and investigation

Time	Duration	Topic / activity	Facilitation content	Slides	Method	Evidence
09:00–09:30	30 min	Administration	Welcome, attendance, profiles, outcomes, TSC, course and assessment briefing	1–19	Facilitation + peer introduction	Baseline and expectations
09:30–11:00	90 min	Topic 01 + Lab 01	Exploring DevOps; shared ownership; problem lifecycle; lifecycle baseline in R	20–31; Lab 28–30	Concepts + guided R lab	Lifecycle summary and ageing evidence
11:00–12:30	90 min	Topic 02 + Lab 02	Three Ways, Theory of Constraints, safe-to-fail and automated signal detection	32–43; Lab 40–42	Process maps + R lab	Severity-ranked actionable alerts
12:30–13:30	60 min	Lunch	Lunch break	—	—	—
13:30–15:00	90 min	Topic 03 + Lab 03	CI/CD, delivery strategies, SRE, DevSecOps, ChatOps, Kanban and	44–55; Lab 52–54	Comparison + case evidence	Pareto and 5 Whys evidence

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			Pareto investigation			
15:00–17:00	120 min	Topic 04 + Lab 04	Agile, Lean, ITSM, safety culture, diagnosis methods and transparent prioritisation	56–67; Lab 64–66	Facilitation + group discussion + R lab	Ranked problem backlog and trade-off
17:00–18:00	60 min	Day 1 synthesis	Connect detection, investigation and priority into one controlled problem record; retrieval practice	56–67	Peer explanation + reflection	Day 1 synthesis checklist

Day 1 total excluding lunch: 480 minutes (8 hours).

Day 2 — Culture, automation, measurement, leadership and assessment

Time	Duration	Topic / activity	Facilitation content	Slides	Method	Evidence
09:00–10:30	90 min	Topic 05 + Lab 05	CALMS, Westrum culture, cultural debt, team models and pipeline-flow diagnosis	68–79; Lab 76–78	Concepts + R lab	Constraint and cultural improvement
10:30–12:00	90 min	Topic 06 + Lab 06	Automation, cloud, containers, microservices, IaC, AIOps, toolchains and proportional guardrails	80–91; Lab 88–90	Architecture comparisons + R lab	Reliability and security policy
12:00–13:00	60 min	Lunch	Lunch break	—	—	—
13:00–14:30	90 min	Topic 07 + Lab 07	Current DORA metrics, SLI/SLO/SLA, reporting and controlled problem documentation	92–103; Lab 100–102	Native chart interpretation + R lab	Five metrics and tracked actions
14:30–16:00	90 min	Topic 08 + Lab 08	Enterprise adoption, leadership, sharing and capstone verified recommendation	104–115; Lab 112–114	Case synthesis + briefing	Problem record and three recommendations
16:00–17:00	60 min	Written Assessment	Five open-ended WA-	116–end wrap/assess	Individual open-book	Candidate WA paper

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			SAQ questions covering K1–K5	ment slides	assessment	
17:00–18:00	60 min	Case Study	Three connected tasks covering A1–A6	116–end wrap/assessment slides	Individual open-book assessment	Candidate CS paper

Day 2 total excluding lunch: 480 minutes (8 hours).

Slide and Lab Mapping

Topic	LO	K/A	Deck slides	Aligned lab
Topic 01 — Exploring DevOps	LO1	K1 · A1	20–31	Lab 01: Build an Incident-to-Problem Lifecycle Baseline in R (slides 28–30)
Topic 02 — Core DevOps Principles	LO1	K2 · A2	32–43	Lab 02: Detect Actionable Service Signals in R (slides 40–42)
Topic 03 — Key DevOps Practices	LO1	K2 · A2	44–55	Lab 03: Investigate Recurring Failure Causes with Pareto Analysis (slides 52–54)
Topic 04 — Business and Technology Frameworks	LO2	K3 · A3	56–67	Lab 04: Prioritise Problems by Severity, Frequency and Impact (slides 64–66)
Topic 05 — Culture, Behaviours and Operating Models	LO2	K4 · A4	68–79	Lab 05: Diagnose Pipeline Flow and Cultural Debt (slides 76–78)
Topic 06 — Automation and DevOps Toolchains	LO2	K4 · A4	80–91	Lab 06: Balance Reliability, Security and Delivery Guardrails (slides 88–90)
Topic 07 — Measurement, Metrics and Reporting	LO3	K5 · A6	92–103	Lab 07: Build a DORA and Problem-Management Evidence Dashboard (slides 100–102)
Topic 08 — Sharing, Shadowing and Evolving	LO3	K5 · A5	104–115	Lab 08: Capstone — Produce a Verified DevOps Problem Recommendation (slides 112–114)

Facilitator Preparation

- Verify the exact current PPT/PDF, Learner Guide, Lesson Plan and lab package all show v4.0.

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- Run every R script from its own folder and confirm all eight LAB NN PASS markers before class.
- Confirm the LMS-TMS download links and assessment candidate-paper links resolve; do not expose answer keys.
- Use the slides for concepts and discussion. Direct learners to the Learner Guide and lab README for detailed procedures.
- Use the synthetic Northstar Retail storyline; do not request production data or credentials.
- Allow short comfort breaks within facilitated time without reducing the approved instructional duration.
- Before assessment, repeat the open-book boundary, individual-work rule, submission path, C/NYC decision and appeal entitlement.

Assessment Traceability

Criterion	Assessment item	Taught in	Practised in
K1	WA Q1	Topics 01–02	Lab 01
K2	WA Q2	Topics 02–03	Labs 02–03
K3	WA Q3	Topic 04	Labs 03–04
K4	WA Q4	Topics 05–06	Labs 04–06
K5	WA Q5	Topics 07–08	Labs 07–08
A1, A2	CS Task 1	Topics 01–03	Labs 01–02 and 08
A3, A4	CS Task 2	Topics 04–06	Labs 03–06 and 08
A5, A6	CS Task 3	Topics 07–08	Labs 07–08

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